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* What is HTTP?

- HTTP (HyperText Transfer Protocol) is the protocol used to transfer web pages. It operates at Layer 7 (Application Layer) of the OSI model.
- Port 80 → HTTP (unencrypted)
- Port 443 → HTTPS (encrypted with TLS)

* HTTP Request Structure

```
GET /index.html HTTP/1.1
Host: www.example.com
User-Agent: Mozilla/5.0
Accept: text/html
Connection: keep-alive
```

- Meaning

GET → HTTP Method

/index.html → Path / resource requested

HTTP/1.1 → Protocol version

Host → Which website

User-Agent → Browser / OS info

* HTTP Response Structure

```
HTTP/1.1 200 OK
```

Content - Type : text/html
Content - Length : 1234

<html> ... </html>

* HTTP Methods

Method	Purpose	DFIR Relevance
GET	Request Data	Normal browsing
POST	Send Data to Server	Login, form, submit, data enfiltration
PUT	Upload / Update	File upload attacks
DELETE	Delete resource	Destructive actions

* HTTP Status Codes

Code	Meaning	DFIR Note
200	OK - Success	Normal
301/302	Redirect	Malware Redirects
403	Forbidden	Access attempt
404	Not Found	Scanning / probing
500	Server Error	Exploitation attempt

Note: POST requests are critical in DFIR — malware uses POST to send stolen data to C2 servers.

* HTTP vs HTTPS

Feature	HTTP	HTTPS
Port	80	443
Encryption	None	TLS Encrypted
Visibility	Full content visible	Only metadata visible
Wireshark Filter	http	~~https~~ tls
Can See headers?	Yes - Everything	No — Encrypted
Can See Domain?	Yes	Yes — via SNI

* What is TLS?

TLS (Transport Layer Security) encrypts HTTP traffic to create HTTPS. Even though the content is encrypted, forensics investigators can still extract information:

TLS Handshake Sequence:

1) Client Hello → client says "I want

to connect securely".

2) Server Hello → Server agrees, sends certificate.

3) Key Exchange → both sides agree on encryption keys

4) Encrypted data flows.

* SNI — The Key Forensics Technique

- SNI (Server Name Indication) is a field in the TLS Client Hello packet that reveals the destination domain name — even when traffic is fully encrypted.

TLS Client Hello
└ Extension: server_name
  └ Server Name: www.google.com

* This is visible *

* Why this matters for DFIR:

- Even encrypted malware traffic reveals its C2 domain via SNI

- You don't need to decrypt the traffic

- The filter: tls.handshake.extensions_server_name

* HTTP in Wireshark — key filters

Filter	What it shows
http	All HTTP traffic
http.request.method == "GET"	Only GET requests
http.request.method == "POST"	Only POST requests ← check these!
tLS	All encrypted HTTPS traffic
tls.handshake.type == 1	TLS Client Hello only
tls.handshake.extensions_server_name	SNI field visible

http

No.	Time	Source	Destination	Protocol	Lengtl	Info
152	3.858850100		103.88.220.57	HTTP	303	GET /c/msdownload/update/others/2026/04/45088676_
157	3.867100600		192.168.0.142	HTTP	490	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
159	3.867812400		103.88.220.57	HTTP	303	GET /c/msdownload/update/others/2026/04/45088102_
164	3.875325500		192.168.0.142	HTTP	445	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
166	3.875997800		103.88.220.57	HTTP	303	GET /c/msdownload/update/others/2026/04/45088095_
172	3.887299100		192.168.0.142	HTTP	461	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
174	3.887879600		103.88.220.57	HTTP	303	GET /c/msdownload/update/others/2026/04/45088112_
179	3.897037400		192.168.0.142	HTTP	427	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
181	3.897659200		103.88.220.57	HTTP	303	GET /c/msdownload/update/others/2026/04/45088101_
187	3.906349100		192.168.0.142	HTTP	439	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
189	3.908373300		103.88.220.57	HTTP	303	GET /d/msdownload/update/others/2026/04/45086894_
193	3.916563800		192.168.0.142	HTTP	462	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
195	3.917193200		103.88.220.57	HTTP	303	GET /d/msdownload/update/others/2026/04/45086912_
199	3.926938800		192.168.0.142	HTTP	443	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
201	3.927564600		103.88.220.57	HTTP	303	GET /d/msdownload/update/others/2026/04/45086911_
207	3.939469100		192.168.0.142	HTTP	449	HTTP/1.1 200 OK (application/vnd.ms-cab-compress
209	3.940102700		103.88.220.57	HTTP	303	GET /d/msdownload/update/others/2026/04/45087422_

```
> Frame 152: Packet
> Ethernet II, Src:
> Internet Protocol
> Transmission Conti
v Hypertext Transfer Protocol
  > GET /c/msdownload/update/others/2026/04/45088676_7c3f4e312356d
    Connection: Keep-Alive\r\n
    Accept: */*\r\n
    User-Agent: Windows-Update-Agent/10.0.10011.16384 Client-Protc
    Host: download.windowsupdate.com\r\n
    \r\n
    [Response in frame: 157]
    [Full request URI: http://download.windowsupdate.com/c/msdownl
```

```
0050  74 68 65 72 73 2f 32 30  32 36 2f 30 34 2f 34 35   thers/20 2
0060  30 38 38 36 37 36 5f 37  63 33 66 34 65 33 31 32   088676_7 c
0070  33 35 36 64 37 61 62 32  64 61 37 37 63 33 35 38   356d7ab2 d
0080  32 35 61 65 30 39 36 37  31 30 66 32 38 64 66 2e   25ae0967 1
0090  63 61 62 20 48 54 54 50  2f 31 2e 31 0d 0a 43 6f   cab HTTP /
00a0  6e 6e 65 63 74 69 6f 6e  3a 20 4b 65 65 70 2d 41   nnection :
00b0  6c 69 76 65 0d 0a 41 63  63 65 70 74 3a 20 2a 2f   live··Ac c
00c0  2a 0d 0a 55 73 65 72 2d  41 67 65 6e 74 3a 20 57   *··User- A
00d0  69 6e 64 6f 77 73 2d 55  70 64 61 74 65 2d 41 67   indows-U p
00e0  65 6e 74 2f 31 30 2e 30  2e 31 30 30 31 31 2e 31   ent/10.0 .
```

tls

No.	Time	Source	Destination	Protocol	Lengtl	Info
52	1.606628200		135.18.128.208	TLSv1.2	1494	Application Data
64	1.813588300		135.18.128.208	TLSv1.2	101	Application Data
66	1.992196200		104.16.102.112	TLSv1.2	96	Application Data
82	2.036183200		192.168.0.142	TLSv1.2	704	Application Data, Application Data
84	2.086768400		192.168.0.142	TLSv1.2	92	Application Data
85	2.087227200		104.16.102.112	TLSv1.2	96	Application Data
87	2.190205400		192.168.0.142	TLSv1.2	92	Application Data
98	2.239778000		192.168.0.142	TLSv1.2	4374	Application Data, Application Data
109	2.243198700		192.168.0.142	TLSv1.2	1494	Application Data, Application Data
123	2.447966700		192.168.0.142	TLSv1.2	1494	[TCP Out-Of-Order] , Application Data, Application Data
128	2.449144000		192.168.0.142	TLSv1.2	1494	[TCP Out-Of-Order] , Application Data
134	3.015513700		18.97.36.3	TLSv1.2	293	Application Data
135	3.307321000		192.168.0.142	TLSv1.2	171	Application Data
138	3.325694000		135.18.128.208	TLSv1.2	198	Application Data
140	3.325805300		135.18.128.208	TLSv1.2	1118	Application Data
145	3.535561800		192.168.0.142	TLSv1.2	889	Application Data
415	8.453370200		18.214.245.199	TLSv1.2	569	Client Hello (SNI=httpbin.org)
416	8.520631800		192.168.0.142	TLSv1.2	92	Application Data
417	8.521241400		104.16.102.112	TLSv1.2	96	Application Data
424	8.713359000		18.214.245.199	TLSv1.2	505	Client Hello (SNI=httpbin.org)
427	8.720633100		192.168.0.142	TLSv1.2	210	Server Hello, Change Cipher Spec, Encrypted Handshake Mes
428	8.721020400		18.214.245.199	TLSv1.2	105	Change Cipher Spec, Encrypted Handshake Message
429	8.721407400		18.214.245.199	TLSv1.2	153	Application Data
430	8.721559400		18.214.245.199	TLSv1.2	593	Application Data
432	8.982720100		192.168.0.142	TLSv1.2	210	Server Hello, Change Cipher Spec, Encrypted Handshake Mes

```
v Transport Layer Security
  [Stream index: 2]
  v TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Content Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 16408
    Encrypted Application Data [...]: 0000000000000088c9e56d6a041624c7a22722b4bc17d80d0d9ed5281b7f95b9bb0d4db6b3058533ab0ac2b492d0
    [Application Data Protocol: Hypertext Transfer Protocol]
v Transport Layer Security
  [Stream index: 2]
  v TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
    Content Type: Application Data (23)
    Version: TLS 1.2 (0x0303)
    Length: 33
    Encrypted Application Data: 00000000000000009085586807f083f108d29d861bcc3a2abcba78904a1d30c338a
    [Application Data Protocol: Hypertext Transfer Protocol]
  TLS segment data (218 bytes)
```